SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	**AKTCHAYA A**	**Reg. No.:**	**192424096**
Date:	**03/08/2026**	**Duration:**	60 minutes

Code of Conduct

I AKTCHAYA A certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________________________

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25

AGILE SPRINT PLANNING AND USER STORY WORKSHOP

UNIVERSITY ACADEMIC MANAGEMENT AND STUDENT SUCCESS PLATFORM

1. Project Vision and Stakeholders

- **Project Vision**

  To develop a secure, scalable, and centralized University Academic Management and Student Success Platform that efficiently manages academic operations, improves communication among stakeholders, identifies academically at risk students, and enhances overall student success through data-driven decision making.

- **Project Objectives**

  - Manage complete student academic records.
  - Support online student admissions and course registration.
  - Monitor attendance and academic performance.
  - Manage examinations and grading.
  - Notify students and faculty about academic activities.
  - Generate academic and institutional reports.
  - Identify academically at-risk students.
  - Improve student engagement.
  - Support academic planning and decision-making.
  - Ensure scalability, security, and high system availability.

• **Key Stakeholders**

Stakeholder	Role
Students	Register courses, view attendance, grades,
Faculty Members	notifications Manage attendance, marks, assessments,
Academic Advisors	course materials Monitor student progress and provide
Department Heads	guidance Monitor departmental performance and
University	faculty activities Manage overall academic operations
Administrators	and reports Conduct examinations and publish results
Examination Cell	Manage admissions and student enrollment
Admission Office	Maintain platform security and availability
IT/System Administrator	Monitor student progress
Parents/Guardians	Review institutional performance and planning

2. Epics and User Stories

Epic 1: Student Management

User Story 1

As a student, I want to register into the university system, so that I can access academic services.

User Story 2

As a student, I want to update my profile, so that my personal information remains accurate.

User Story 3

As an administrator, I want to maintain student records, so that academic information is centralized.

Epic 2: Course Registration

User Story 4

As a student, I want to register for courses online, so that I can enroll without paperwork.

User Story 5

As faculty, I want to approve course registrations, so that eligible students are enrolled.

Epic 3: Attendance Management

User Story 6

As faculty, I want to record attendance, so that student attendance is tracked accurately.

User Story 7

As a student, I want to view my attendance, so that I know my attendance percentage.

Epic 4: Examination & Assessment

User Story 8

As faculty, I want to upload marks, so that student performance is recorded.

User Story 9

As students, I want to view examination results, so that I know my academic performance.

Epic 5: Student Success Monitoring

User Story 10

As an academic advisor, I want to identify academically at-risk students, so that timely support can be provided.

User Story 11

As a student, I want to receive academic alerts, so that I can improve my performance.

Epic 6: Reports & Dashboard

User Story 12

As an administrator, I want to generate academic reports, so that institutional performance can be analyzed.

User Story 13

As department heads, I want to monitor faculty performance, so that departmental quality improves.

3. Acceptance Criteria (Given–When–Then)

- **User Story 1**
  - **Student Registration**
    - **Given** the student is on the registration page
    - **When** valid details are entered
    - **Then** the student account should be created successfully.
- **User Story 4**
  - **Course Registration**
    - **Given** the student has logged in
    - **When** eligible courses are selected
    - **Then** the selected courses should be registered successfully.
- **User Story 6**
  - **Attendance Recording**
    - **Given** faculty is logged in
    - **When** attendance is submitted
    - **Then** attendance should be stored correctly.
- **User Story 8**
  - **Upload Marks**
    - **Given** faculty has completed assessment
    - **When** marks are entered
    - **Then** student grades should be updated.
- **User Story 10**
  - **At-Risk Student Identification**
    - **Given** attendance and marks are below university policy
    - **When** academic analysis is performed

        - **Then** the student should be flagged as academically at-risk.
- **User Story 12**
    - **Academic Reports**
        - **Given** administrator selects report criteria
        - **When** report generation is requested
        - **Then** the system should generate downloadable reports.

4. Product Backlog Prioritization

Priority	Product Backlog Item	Business Value	Priority Level
1	Student Registration	Very High	Must Have
2	Login & Authentication	Very High	Must Have
3	Student Record Management	Very High	Must Have
4	Course Registration	Very High	Must Have
5	Attendance Management	High	Must Have
6	Examination Management	High	Should Have
7	Grade Management	High	Should Have
8	Academic Notifications	Medium	Should Have
9	At-Risk Student Detection	High	Should Have
10	Dashboard & Reports	Medium	Could Have
11	Faculty Activity Tracking	Medium	Could Have
12	Parent Portal	Low	Won't Have (Sprint 1)

5. Sprint Backlog (Sprint 1)

- **Sprint Duration: 2 Weeks**

User Story	Development Tasks
Student Registration	Design registration page, validate input, store data
Login System	Create login page, authentication, password encryption
Student Records	Design database, CRUD operations
Course Registration	Create course selection page, enrollment validation
Attendance Module	Faculty attendance page, attendance database

- **Actionable Tasks**
    - Requirement analysis
    - Database design
    - UI wireframe creation
    - Registration module
    - Login module
    - Student dashboard
    - Course registration page
    - Attendance module
    - Unit testing
    - Integration testing
    - Documentation

6. Story Point Estimation (Fibonacci)

User Story	Story Points	Justification
Student Registration	**3**	Simple form with validation and database storage
Login & Authentication	**5**	Requires authentication and security
Student Record Management	**8**	Multiple CRUD operations and database integration
Course Registration	**8**	Includes enrollment rules and validations
Attendance Management	**5**	Moderate complexity with faculty interaction

- **Estimation Technique Used**
  - **Fibonacci Sequence (1, 2, 3, 5, 8, 13, 21)**
    - **Justification**
      - **3 Points:** Low complexity with minimal logic.
      - **5 Points:** Moderate effort involving security and validation.
      - **8 Points:** Higher complexity due to multiple workflows, business rules, and database interactions.

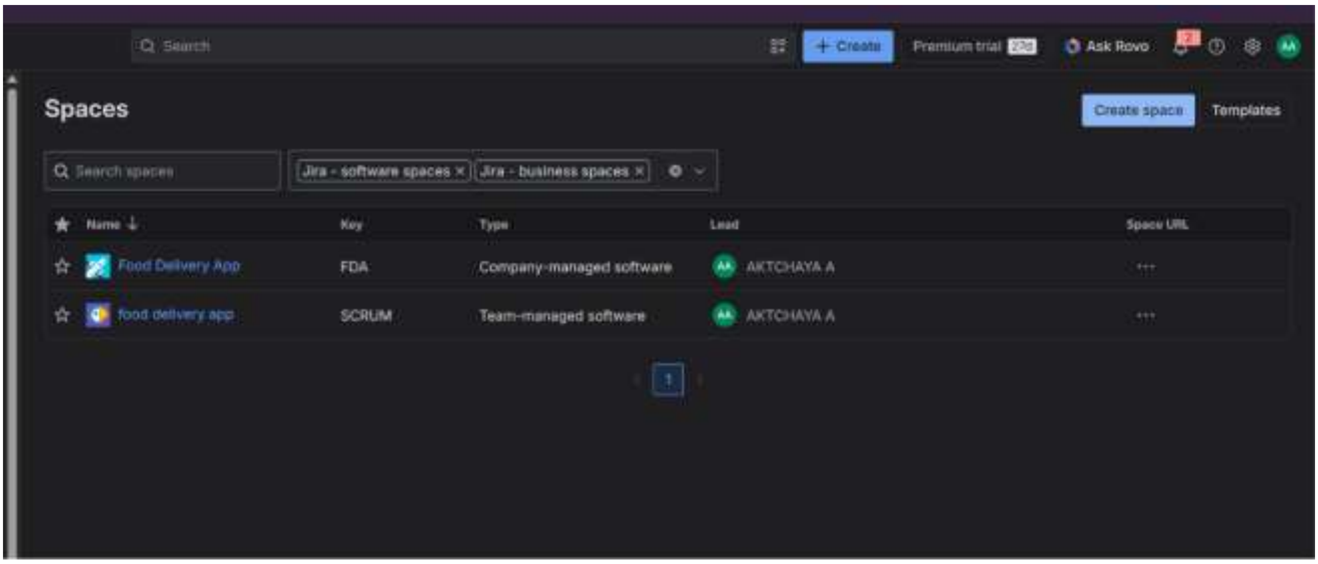
7. Sprint Goal and Expected Deliverables

- **Sprint Goal**
  - Develop the core functionalities required for student onboarding and academic management, including user authentication, student record management, course registration, and attendance tracking, providing a strong foundation for future academic modules.
- **Expected Deliverables**
  - **Functional Features**
    - Student Registration Module
    - Secure Login & Authentication
    - Student Profile Management
    - Student Record Database

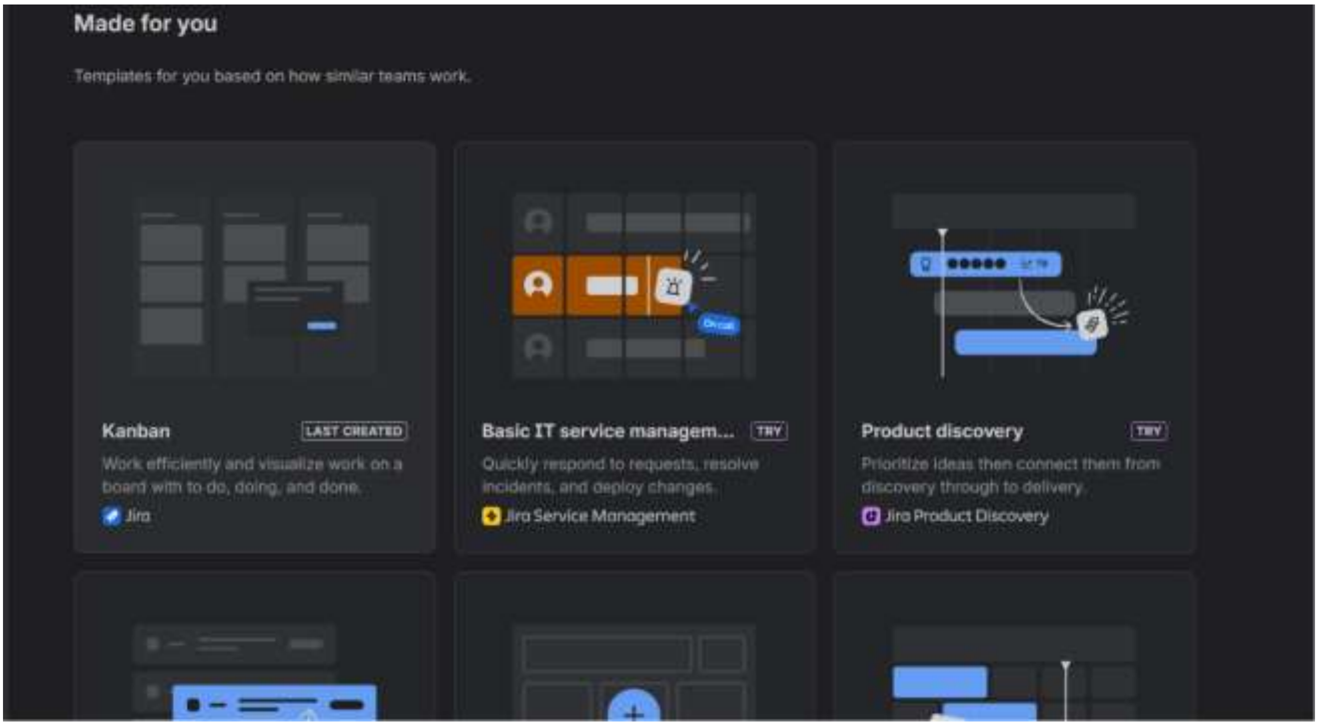
        - Course Registration Module
        - Attendance Management Module
    - **Technical Deliverables**
        - Database schema
        - Responsive user interface
        - Backend APIs
        - User authentication
        - Unit-tested core modules
        - Sprint documentation
    - **Sprint Outcome**
        - At the end of Sprint 1, the platform will allow:
        - Students to register and log in.
        - Administrators to manage student records.
        - Students to enroll in courses.
        - Faculty to record attendance.
        - The system to securely store academic data, providing a foundation for later features such as examinations, performance tracking, notifications, reporting, and at-risk student identification.

8.Screenshot of implementation:

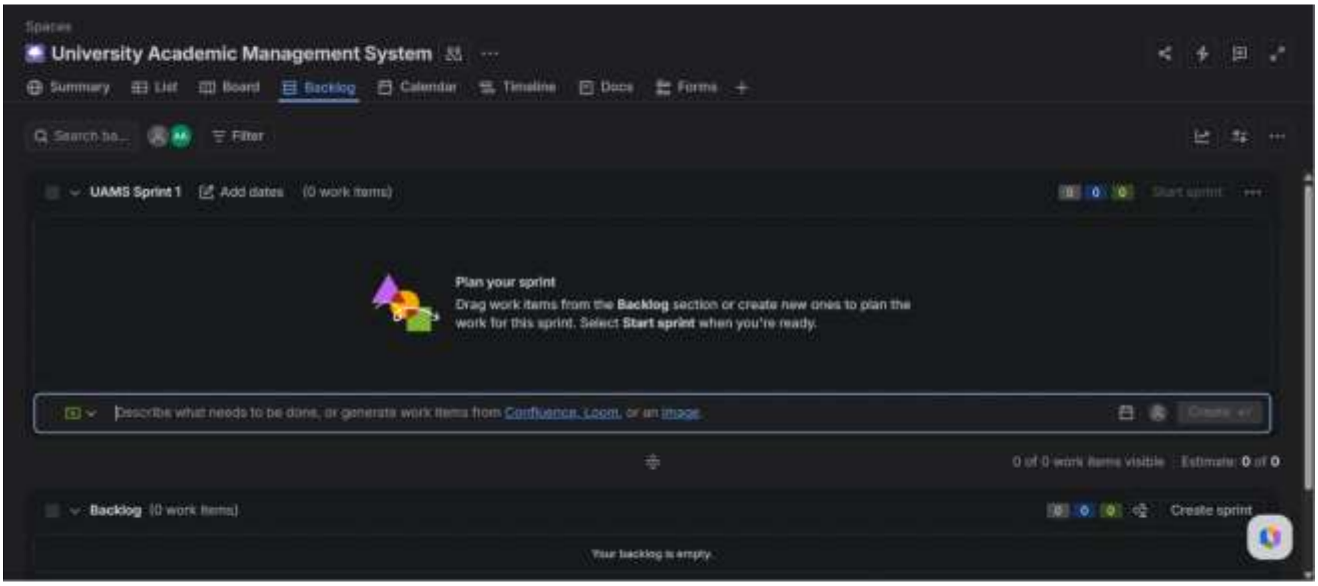
- **Jira Home Page**



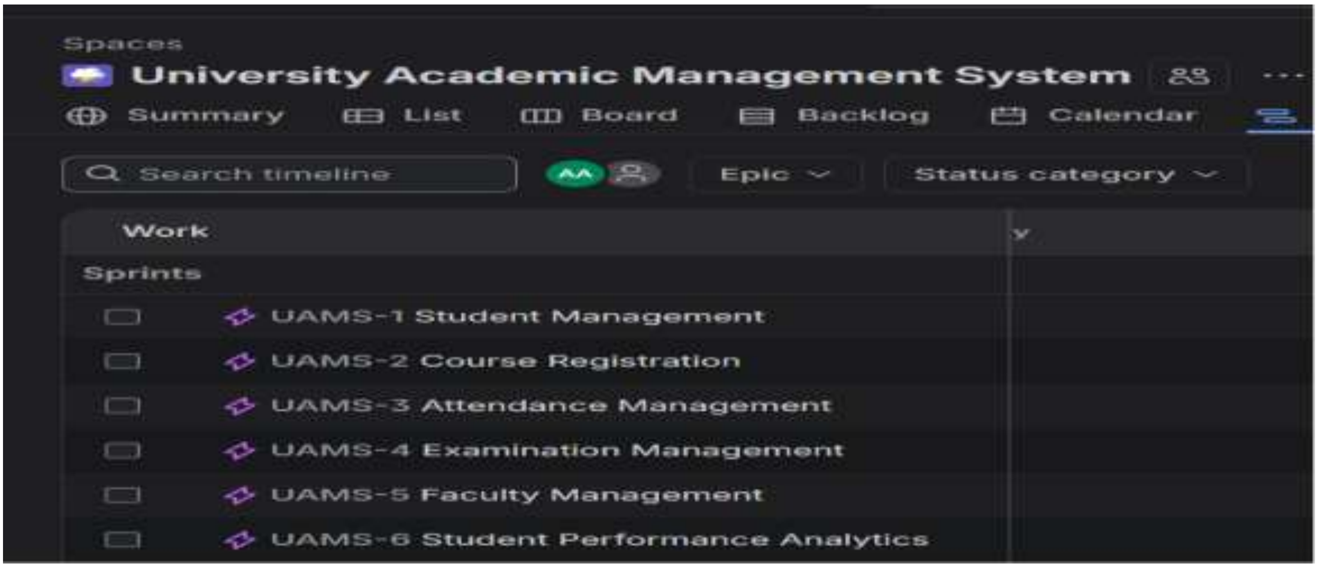
- **Create Project**



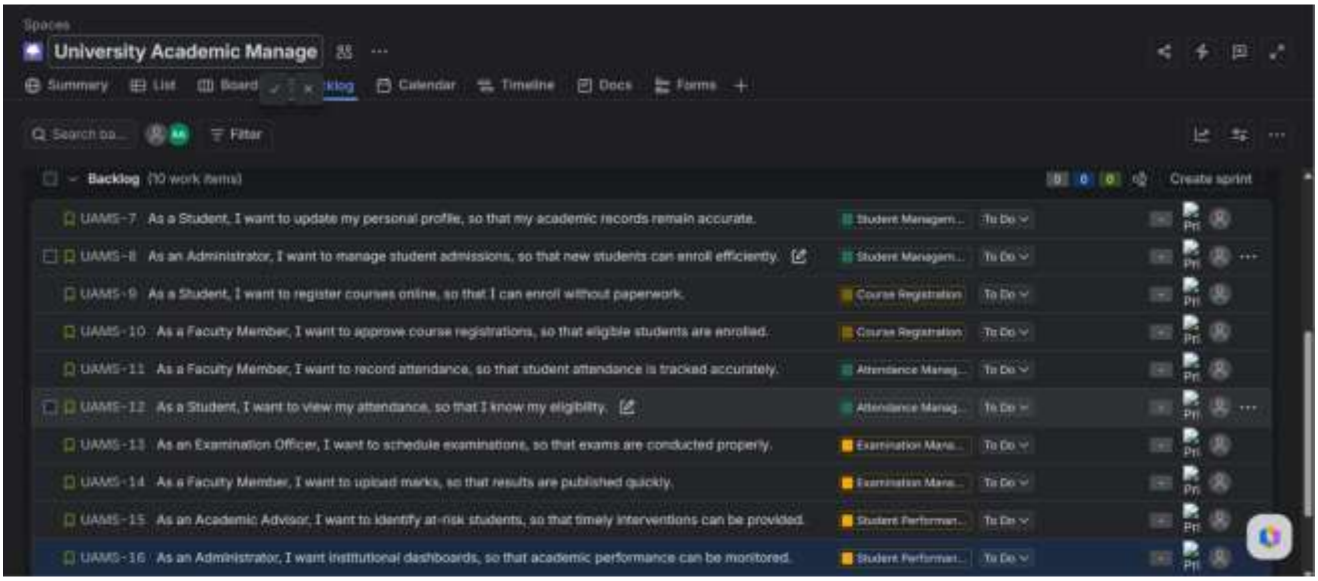
- **Scrum Project created**



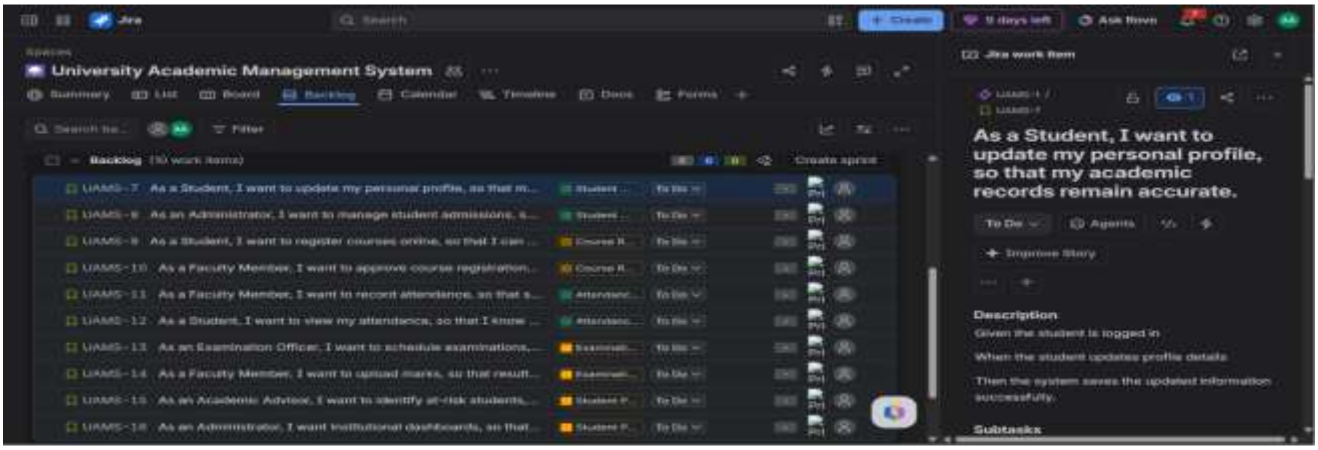
- **Epic List**



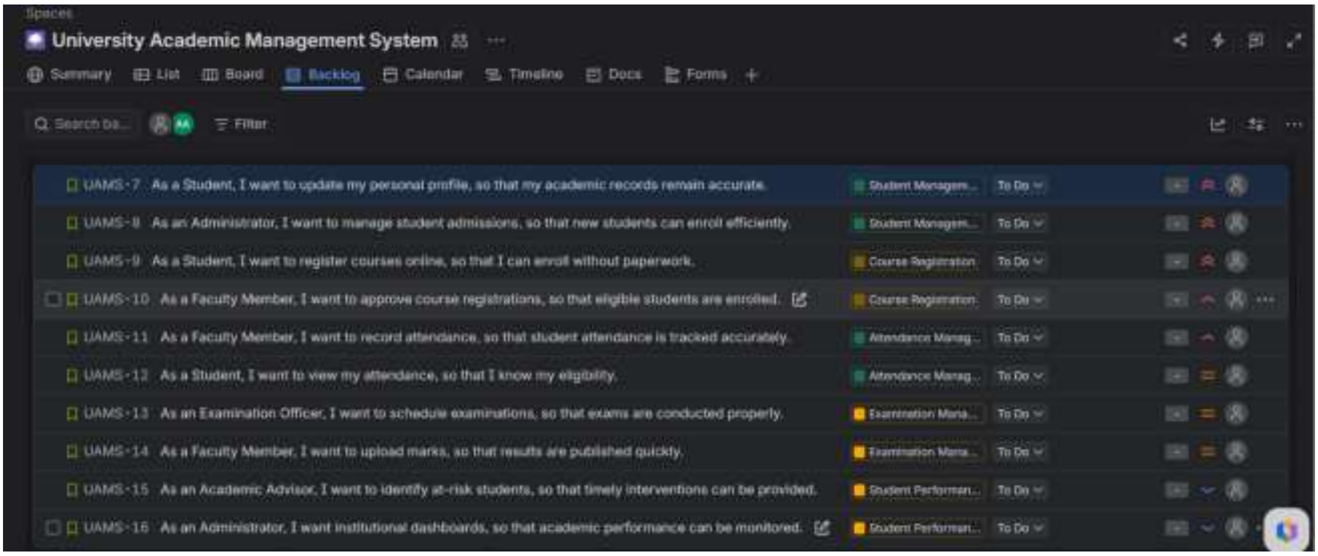
- **User Stories in Backlog**



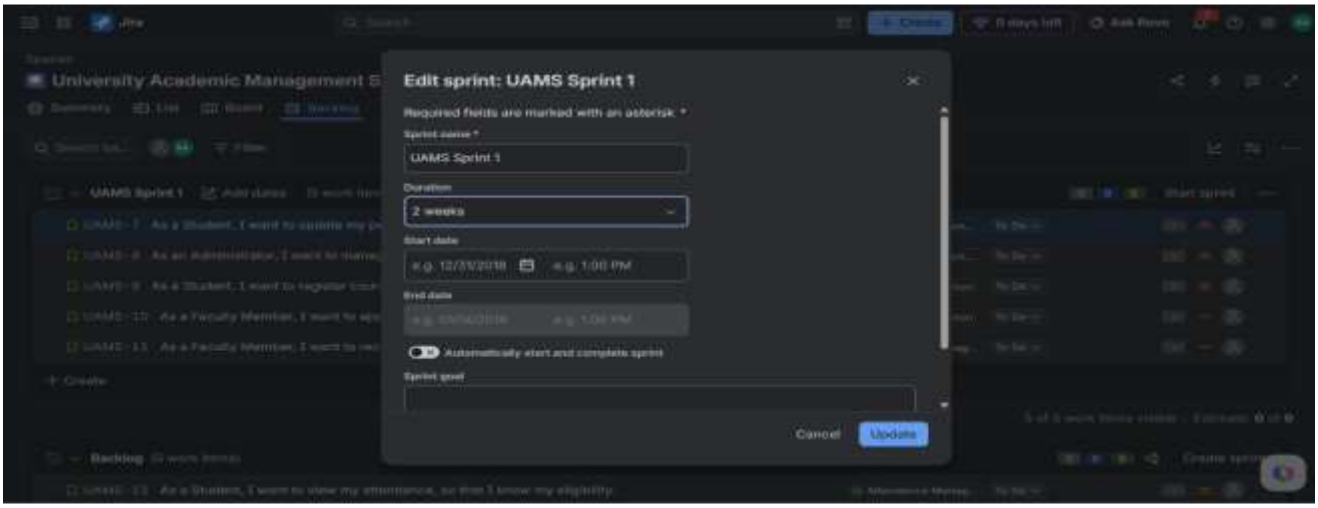
- **Acceptance Criteria for a story**



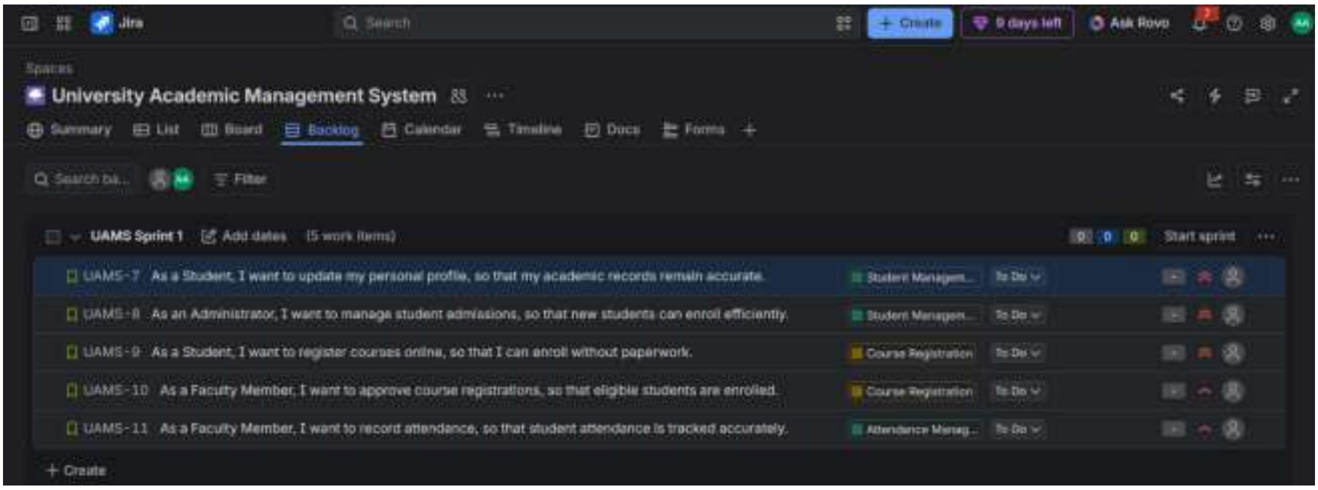
- **Prioritize the Backlog**



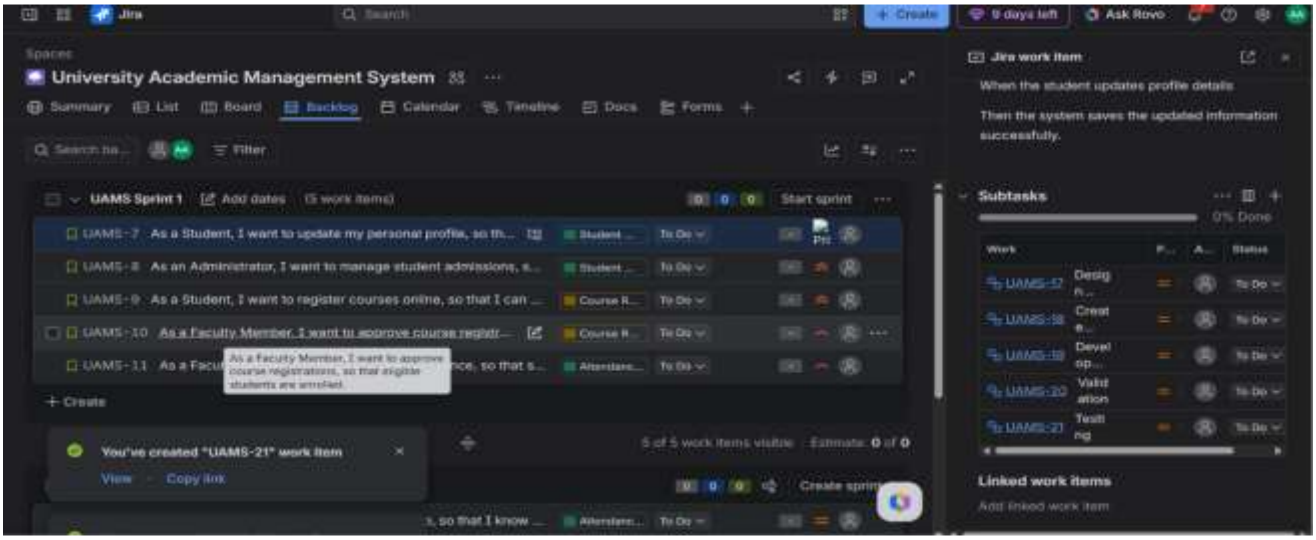
- **Sprint Creation**



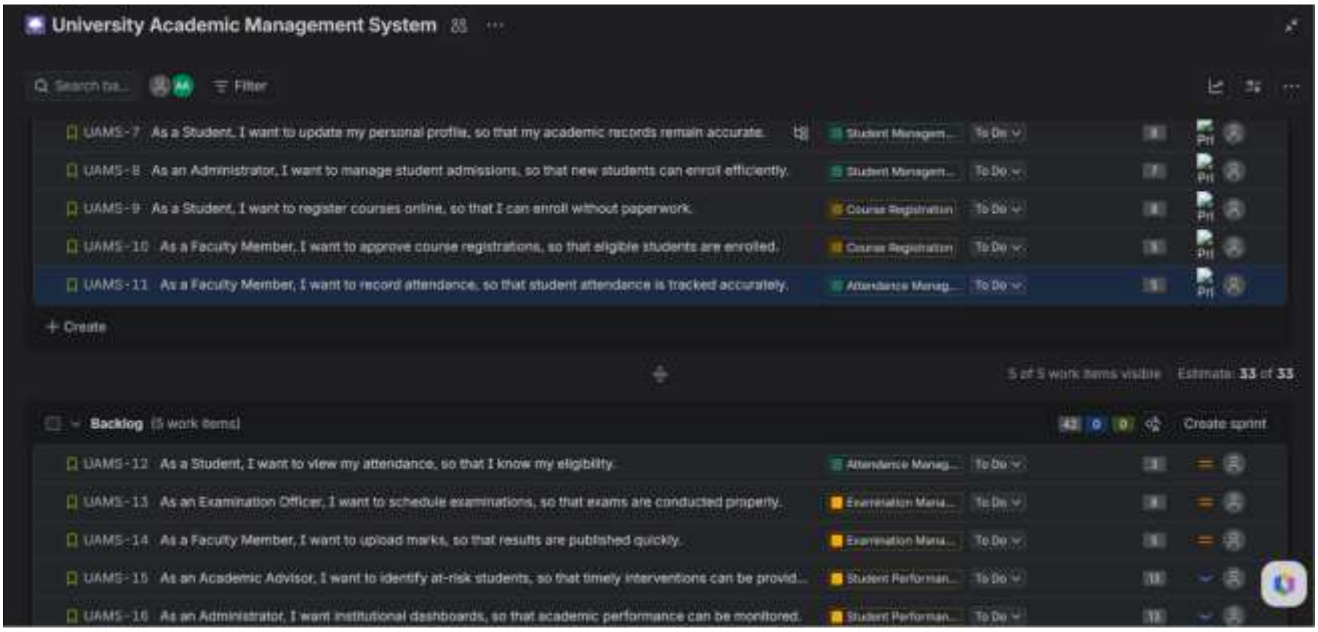
- **Stories added to Sprint**



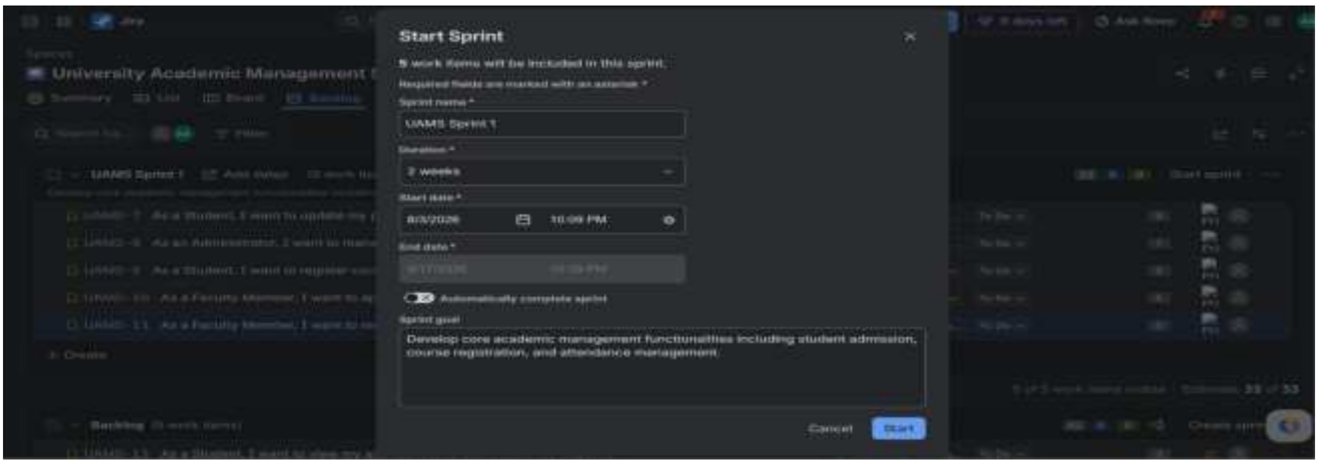
- **Task under a story**



- **Story Points Assigned**



- **Sprint Goal dialog**



- **Active Scrum Board**